```
            "flood_stage": "90%",
            "high": "85%"
          }
        }
      }
    }
  }
}
...
```

We can also provide all these settings simultaneously. I'll leave it up to you to experiment (or see the book's files for the solution).

Unresponsive server calls are always a problem in distributed systems. Users sometimes have to wait longer than needed only to realize that their call failed, but the server didn't return quickly enough to notify them. To avoid such error conditions, software systems implement circuit breakers. Elasticsearch has them too, as we discuss in the following section.

15.4 Circuit breakers

In distributed architectures and applications, failure of remote calls from nonresponsive services is a given, although it shouldn't be taken for granted. It is a common problem in a microservices world for a client to wait longer than usual only to receive an error. Fortunately, there is a circuit-breaker pattern to alleviate this problem.

A *circuit breaker* is a fallback method that is triggered when a response crosses a threshold time due to server-side problems such as being out of memory, resource locks, and so forth. It's like waiting a long time in a queue for a new iPhone, only to be disappointed when the store runs out—but they give you a gift certificate.

Elasticsearch is no different. It's a distributed application that's expected to throw errors and exceptions for many reasons. Elasticsearch implements circuit breakers to counteract problems that hamper a client's progress. It throws meaningful errors to the client if the circuit breakers are triggered.

Elasticsearch has six circuit breakers for various scenarios, including a parent-level catch-all circuit breaker. For example, if overall memory on the node increases due to the overall consumption of the current inflight requests, the inflight-requests circuit breaker kicks in to save the node from failure. The following query uses the nodes API to fetch the current memory limits set on various circuit breakers.

Listing 15.17 Fetching circuit breaker memory settings

```
GET _nodes/stats/breaker
```

Elasticsearch throws errors when the memory required to carry out an operation is inadequate (or action), and it triggers the circuit breakers. Clients get an "Out of Memory" error instantly from these circuit breakers. Table 15.1 lists the various types of circuit breakers, their limits, and the associated properties.